

Hamza Bouktitiya

Artificial Intelligence Engineer

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📍 Kenitra, Morocco

🌐 LinkedIn 🐙 GitHub 📁 Portfolio

Education

- **PhD Candidate in Artificial Intelligence** *2024 – Present*
Ibn Tofail University, Kenitra, Morocco. Thesis: Textual description and automatic classification of video sequences by deep learning. Supervised by: Professor Housni Khalid
- **Master in Artificial Intelligence and Virtual Reality** *2022 – 2024*
Ibn Tofail University, Kenitra, Morocco
- **Bachelor in Mathematical and Computer Science** *2018 – 2022*
Ibn Tofail University, Kenitra, Morocco
- **Baccalaureate in Mathematical Sciences** *2017 – 2018*
Sidi Aissa High School, Souk El Arbaa Du Gharbe

Professional Experience

- **Junior AI Engineer** *April 2025 – October 2025*
OBYSTECH SOLUTIONS, Rabat
Developed a unified AI agent platform serving HR, Finance, Management, and IT departments (solo project). Front-end: multi-chat interface with collection selector, RAG toggle per chat, team workspaces with public/private sharing, cursor-based pagination. Back-end: RAG pipeline with document ingestion (PDF/docs/txt/md), HF embeddings, hybrid/vector search with MMR re-ranking, LangGraph workflows with PostgreSQL-managed memory, MCP integration for external tools. Deployment: multi-service Docker Compose stack, VPS deployment with reverse proxy/SSL, zero-downtime restarts. Technologies: Next.js, React, Tailwind, FastAPI, LangChain, LangGraph, MCP, Ollama, PostgreSQL, ChromaDB, Docker, JWT/OAuth2
- **AI Intern** *February 2024 – August 2024*
Verve Technologie, Casablanca
Built a forecasting system using LSTM to predict energy consumption. Set up image generation endpoints with Stable Diffusion 1.5. Developed a RAG system to generate text from Wikipedia and PDF. Added JWT authentication and user management. Technologies: LSTM, LangChain, FastAPI, Stable Diffusion, JWT
- **Web Development Intern** *March 2022 – June 2022*
CS Department, Ibn Tofail University, Kenitra
Created a web application for gym management (clients and administrators). Technologies: PHP, JavaScript, Bootstrap, CSS, MySQL

Technical Skills

AI & Machine Learning: PyTorch, PyTorch3D, TensorFlow, LangChain, LangGraph, CrewAI **Back-end:** Django, FastAPI, SQL, MySQL, PostgreSQL, PL/SQL **Front-end & Design:** HTML, CSS, Bootstrap, Next.js, THREE.JS, LangChain.js **Data Analysis:** Python, Power BI, Pandas, NumPy

Personal Projects

- **Chatbot Odoo** (2024) – AI assistant integrated with Odoo ERP allowing natural language interactions to manage business operations across modules (CRM, Sales, Inventory, HR) using NLP for query interpretation and complex action execution
- **Educational Game - DyslexAI** (February 2024) – Educational game developed during DyslexAI hackathon to help children improve pronunciation and word learning, built with Python and Django framework
- **3D Object Generation** (February 2024) – 3D object reconstruction from 2D images using multi-view generation neural networks and implicit neural representations (NeRF) with PyTorch
- **Health Chatbot - ThinkAI** (August 2024) – Multimodal chatbot developed with Python, Ollama and Streamlit supporting text and image inputs with buffer functionality for continuous interactions
- **Book Recommendation System** (2024) – Recommendation system using collaborative filtering techniques, TF-IDF processing, and machine learning algorithms to analyze user preferences and suggest relevant books with Streamlit interface
- **Darija Q&A System** (December 2023) – NLP system for bilingual question answering in Moroccan darija and English, trained three models: Transformer-based QA model, Seq2Seq for darija-English translation, LSTM for English-darija translation
- **License Plate Recognition** (June 2023) – Custom computer vision model for license plate character detection, generated 1,700 HOG features per character, achieved 98.9% accuracy with KNN and SVM classifiers

Languages & Activities

Languages: Arabic (Native) – English (C1) – French (B2) **Activities:** DyslexAI Hackathon (TOP 3, Al Akhawayn University) – ThinkAI Hackathon (1337 Coding School) – Co-founder Pragnomos Club – ML Presentation (Agora event)